

Stream Functions Assessment and Rapid Index (SFARI) Field Worksheet (Version 1.0)		
Reach ID:	Reach Length:	Date: Assessor(s): Coordinates:
SCORING INSTRUCTIONS		
Function scores are judgment-based evaluations reflecting the stream condition relative to physical or ecological function. Metrics record the logic embedded in the function and can be used to calculate a function score. Each metric has a specific context: W = Watershed, F = Floodplain, C = Channel and Bank. Metrics may be omitted with "NA", or metrics may be added. A minimum of three metrics per function is recommended. Metrics requiring field analysis or desktop analysis are noted with F or D. Assessment involves two steps: 1) Score metrics for agreement with statements: Strongly Agree (SA), Agree (A), Neutral (N), Disagree (D), Strongly Disagree (SD), Not Appl (NA). 2) Score functions using metrics based on the following scale: Functioning (15 to 11), Functioning At-Risk (10 to 6), or Non-Functioning (5 to 0)		
Function	Metrics	Score
HYDROLOGY FUNCTIONS		
Catchment Hydrology: Runoff and infiltration sustain natural flow regime, carry appropriate sediment and nutrients from uplands, and reliably cue spawning/ migration of aquatic life.	Impervious surface area ^{W, D} : Coverage is minimal, preserving near-natural infiltration/runoff timing, consistent with reference levels.	
	Road density ^{W, D} : Road density is low enough to avoid significant runoff or sediment inputs, consistent with minimal watershed impact.	
	Land use change ^{W, D} : <5% land cover shift in ~15–20 years, indicating stable infiltration/runoff consistent with reference conditions.	
	Impoundments ^{W, D} : Flow is near-natural, with no major dams, unless larger dams are normal per reference.	
	Score:	Notes/Other Metrics:
Surface Water Storage: Wetlands and storage features store floodwater, recharge groundwater, sustain baseflow, and provide low-velocity habitat.	Wetland coverage ^{F, D} : Sufficient wetlands/ponds for flood attenuation/baseflow support, unless minimal wetlands are normal per reference.	
	Floodplain water retention ^{F, F} : Moderate floods (~1–5 yr) reach the floodplain, providing water retention/infiltration per regional reference.	
	In-channel ponding/beaver ^{C, F} : Small beaver-type impoundments aid baseflow/habitat; no major fragmentation unless normal per reference.	
	Off-channel storage ^{F, F} : Side channels/beaver ponds/oxbows connect during moderate floods, providing off-channel storage and habitat.	
	Score:	Notes/Other Metrics:
Reach Inflow: Quantity + quality of inflow (tributaries, ditches, and pipes) does not provide harmful peaks or pollution, and supports diverse habitat conditions.	Concentrated flow inputs ^{C, D} : Storm/tile drains, ditches are minimal or absent, preserving near-natural infiltration/runoff timing.	
	Tributary Condition and Impact ^{W/C, F} : Tributaries stable, matching reference (flow, quality, form), with no major flash/pollution surges.	
	Local runoff diversions ^{W, D} : Diversions/returns are minimal, flow remains near natural regime without artificial additions/withdrawals.	
	Road/highway drainage ^{C, F} : Runoff is minimal or well-managed, causing no significant direct inflows or pollutant pulses.	
	Score:	Notes/Other Metrics:
Streamflow Regime: Natural flow regime sustains ecological integrity of river system. Hydrograph shifts (e.g., from hydropeaking, dams, withdrawals) are minimal, with no adverse effect on sediment transport, connectivity, and nutrient or chemical delivery.	Flow permanence ^{C, F/D} : Flow duration category from regional SDAM metrics (ephemeral/ intermittent/ perennial) matches reference. Timing, duration, and frequency of seasonal flows align with reference. No major shift in dryness, baseflow, or mean annual flows, indicating minimal alteration from withdrawals. Streambed vegetation extent and type aligns with expected flow type.	
	Flow statistics ^{C, D} : Quantitative hydrologic statistics (e.g., flow-duration stats, annual flow days, percent of year flow, flashiness, recession, rate-of-change) fall within reference expectation.	
	Channel natural flow regime ^{C, D} : Baseflow to peak floods align with reference condition of magnitude, duration, frequency, and timing of flows indicating minimal hydrologic alteration.	
	Artificial structures and inputs ^{C, D} : No major flood-control or flow-regulating structures alter peak/baseflow. Minimal disruption from pumping, dams, or diversion.	
	Score:	Notes/Other Metrics:

HYDRAULICS FUNCTIONS		
Low Flow and Baseflow Dynamics: Low and base flows, supported by groundwater and hyporheic exchange, stabilizes temperature and water quality, maintains adequate refugia without thermal or DO stress, and concentrates nutrients at beneficial levels overall sustaining aquatic communities.	Low flow velocity ^{C, F+D} : Low-flow velocities, baseflow, and seasonal flow timing match the stream type, reference condition, and historical climate cycles (rain/snowmelt/dry). No unnatural stagnation or dryness compared to reference.	
	Low flow depth ^{C, F+D} : At normal low flow: channel is mostly wetted, exposed channel bed matches the stream's natural wet/dry regime, pools offer sufficient depth for typical fish/invertebrate refuge. Key low/moderate flows align with reference, indicating minimal hydrologic alteration.	
	Longitudinal connectivity ^{C, F} : At normal low flow, longitudinally: velocity changes, depth changes, movement of water and sediment, thalweg depth, and residual pool connectivity match reference or expected condition.	
	Lateral connectivity ^{C, F} : At normal low flow, laterally: width of channel and bed wetted, velocity changes, depth changes, movement of water and sediment, thalweg depth and connectivity to floodplain match reference or expected condition.	
	Score:	Notes/Other Metrics:
High Flow Dynamics: Peak flows reshape channel without chronic instability, redistribute wood and sediment, reset habitats, and supply nutrients to floodplains.	Overbank Flow Frequency ^{F/C, D} : Small floods overtop banks approximately every 1-2 years.	
	Peak Flow Capacity (Morphological Check) ^{C, D} : The channel cross section accommodates bankfull flows (every ~1-2 years) without destructive events.	
	Peak Flow Capacity (Velocity/Shear Stress) ^{C, D/F} : Channel, substrate, and banks accommodate bankfull floods and shear stress without frequent failures or excessive scour/fill, indicating stable capacity.	
	Bed Mobilization Frequency ^{C, D} : Bed sediments mobilize at expected bankfull flood timing (~1-2 year), maintaining a stable but dynamic channel.	
	Score:	Notes/Other Metrics:
Floodplain Connectivity: Frequent overbank flows reduce flood peaks, support riparian vegetation, create off-channel refugia, and extend nutrient processing time.	Floodplain Complexity ^{F, F} : Floodplain/off-channel features stay diverse/connected at moderate floods, reflecting minimal loss.	
	Entrenchment Ratio (ER) ^{C, F} : Channel is not deeply incised; moderate floods access a broad floodplain, matching reference conditions.	
	Channel Condition ^{C, F} : Stable, not heavily dredged/incised, allowing normal overbank flows/meanders unless dredging is expected per reference.	
	Lateral Floodplain Inundation ^{F, D} : 1–2 yr floods routinely access the floodplain, matching reference inundation patterns.	
	Score:	Notes/Other Metrics:
Hyporheic Connectivity: Surface water-groundwater exchange moderates temperature, transforms nutrients, supplies oxygen, and supports aquatic life.	Channel Complexity for Exchange ^{C, F} : Diverse bedforms enhance surface–subsurface exchange, matching reference morphology.	
	Floodplain Permeability ^{F, F} : Floodplain soils allow hyporheic/groundwater exchange, consistent with local reference.	
	Bed Surface Grain Size ^{C, F} : Substrate has coarse particles with minimal fines, enabling interstitial flow per local references.	
	Visible Hyporheic Metrics ^{C, F} : Springs/seeps/subsurface flow show active hyporheic exchange, matching local reference.	
	Score:	Notes/Other Metrics:

GEOMORPHOLOGY FUNCTIONS		
Channel Evolution: Long-term changes in channel size and slope maintain dynamic equilibrium and diverse habitats while interacting with the floodplain.	Channel Evolution ^{C, F} : Channel is stable/recovered (Stage I/VI or 0,1,6,8), not actively degrading, unless partial evolution is expected per reference.	
	Incision Trend (headcuts) ^{C, F} : No active headcuts or fully stabilized ones; no ongoing incision or bank widening.	
	Widening Trend ^{C, F} : Bank erosion/migration rates match natural reference; no excessive widening unless expected per reference.	
	Recovery Indicators ^{C, F} : Channel shows vegetated benches/floodplain formation, indicating recovery from incision/widening.	
	Score:	Notes/Other Metrics:
Channel and Floodplain Dynamics: Balanced bank erosion and deposition supply needed sediment, form bars and undercuts, and sustain diverse habitat without excessive instability. Natural adjustments in sinuosity and side channels redistribute energy and sediment, generate habitat diversity, and drive riparian succession.	Bank Erosion Potential ^{C, F} : Banks are under minimal erosion risk (e.g., rate Very Low/Low hazard by BEHI criteria). Only a small fraction of banks erode, unless higher rates are normal in highly dynamic meanders. Natural/gradual erosion, sufficient vegetation, and quick recovery after highs indicate stability. Banks mostly rely on natural deep-rooted vegetation, unless artificial armoring is normal per reference.	
	Bank Migration and Meander ^{C, D} : Bank migration and meander shifts occur at a natural rate, unless forced straightening or extreme rates are typical for local geology. No abnormal or human-induced channel jumps.	
	Sinuosity ^{C, F} : Sinuosity is appropriate for this stream type, unless lower/higher curvature is normal per reference (e.g., steep vs. meandering).	
	Channel Pattern ^{C/F, F/D} : Channel follows a natural planform and remains consistent with reference: minimal straightening, and no unnatural braiding. Secondary/ side channels match reference, provide extra habitat/capacity unless blocked historically.	
	Score:	Notes/Other Metrics:
Sediment Continuity Balanced sediment supply and transport preserves bed elevations, substrate sizes, spawning/benthic habitats, and supports riparian succession.	Sediment Deposition Patterns ^{C, F} : Deposition matches natural form, no severe bar formation beyond typical references.	
	Channel Degradation (Incision) ^{C, F} : Bed is stable, with no active incision disconnecting floodplain unless minor incision is normal.	
	Fine Sediment Balance ^{C, F} : Fine input aligns with transport, unless higher turbidity/fines are normal per reference (e.g., blackwater).	
	Transport Capacity ^{C, D} : Flow/slope balance incoming sediment, avoiding major bars or deep scour, unless imbalance is normal per reference.	
	Score:	Notes/Other Metrics:
Bed Composition and Bedform Dynamics Heterogeneous mix of grain sizes and bedforms provide benthic habitat niches, support hyporheic flow, and maintain channel roughness. Adequate recruitment and retention of large wood to deflect flow, form pools, stabilize banks, trap carbon, and build complex habitat.	Large Wood Frequency and Diversity ^{C, F} : Large wood frequency is consistent with forested regions, unless minimal per reference (e.g., treeless prairies). Diversity in large wood pieces (rootwads, stable pieces) foster robust habitat in this forested stream.	
	Riparian Wood Recruitment ^{F, F} : The corridor has mature/mid-successional trees ensuring long-term large woody debris input.	
	Substrate Composition ^{C, F} : Classes match reference (no excessive fines or coarsening/armoring beyond typical). Coarse substrate mostly unembedded, unless high fines are normal (wetland streams); embeddedness is healthy.	
	Bedform Diversity ^{C, F} : Riffles, runs, pools (or step-pool), and adequate depths provide habitat complexity matching reference.	
	Score:	Notes/Other Metrics:

PHYSICOCHEMISTRY FUNCTIONS		
Light and Thermal Regime Shade, solar input, and groundwater keep temperature and light within natural ranges for stream metabolism and dissolved oxygen.	Riparian Canopy Cover ^{F, D} : Shade is sufficient for this stream type, matching reference condition or unless naturally open.	
	Stream Temperature ^{C, F} : Stays within acceptable bounds for local aquatic life, rarely exceeding stress thresholds.	
	Algal Growth (light-limited) ^{C, F} : Moderate growth matches local light regime, unless high/low is expected per reference (e.g., blackwater).	
	Thermal Refugia ^{C, F} : Springs or deeper pools temper extremes, unless large daily swings are normal per reference (e.g., arid ephemeral).	
	Score: Notes/Other Metrics:	
Carbon Processing: Organic matter is captured and broken down, fueling food webs, balancing production/ respiration, moderating pH/ redox, and supplying nutrients.	CPOM Retention ^{C, F} : Leaf litter/coarse organic matter are visibly retained, supporting detrital processes per local references.	
	Detritus Decomposition Rate, Shredder/ Detritivore Presence ^{C, F} : Visible fungal colonization/shredding indicates a healthy detrital cycle. Quick checks show invertebrates, indicating healthy detrital processing (if typical).	
	Riparian Corridor Width/Quality ^{F, D} : Sufficient native vegetation width unless narrow per reference (e.g., prairie streams).	
	Algal/Primary Production ^{C, F} : Balanced algae/macrophyte growth, no nuisance blooms, unless moderate coverage is normal per reference.	
	Score: Notes/Other Metrics:	
Nutrient Cycling: Nitrogen and phosphorus transformations feed primary production, regulate excess loads, and store or release nutrients.	Visible Algal Indications (N/P) ^{C, F} : Algae suggests no excessive nutrients vs. reference, unless moderate per reference (e.g., open canopy).	
	N/P Concentrations (surrogates) ^{C, D} : Nutrients near natural/guidelines unless higher levels are normal per reference (e.g., blackwater).	
	Vegetated Riparian Corridor Width ^{F, D} : Sufficient buffer filters nutrients, matching local references.	
	Relative Denitrification Potential (visual) ^{F, F} : Adequate floodplain/wetland pockets for denitrification, consistent with local reference.	
	Score: Notes/Other Metrics:	
Water and Soil Quality: Physical and chemical processes limit pollutants, retain beneficial constituents, and protect ecological and human uses.	Water Clarity/Turbidity ^{C, F} : Appears acceptable, minimal turbidity matching local reference conditions.	
	Dissolved Oxygen ^{C, F} : DO is high enough (>5–6 mg/L) for aquatic life, unless naturally lower in warm/boggy streams.	
	Toxic Pollutants ^{C/F, F} : No visible sheen or strong odor, consistent with an unpolluted stream. No chemical/fecal contamination signs; soils appear natural, no suspicious odors/staining.	
	pH & Specific Conductivity ^{C, F} : Values near ~6.5–8.5/local norms, no extreme shift unless naturally acidic (bog) or alkaline (arid).	
	Score: Notes/Other Metrics:	

BIOLOGY FUNCTIONS		
Habitat Provision: Channel and floodplain structure supply depth, velocity, substrate diversity, and vegetation to support native organisms through all life stages.	In-Stream Habitat Complexity ^{C, F} : Multiple habitat types (riffles/pools/large wood) exist if expected per reference, not unnaturally uniform.	
	Overhanging Vegetation ^{C, F} : Provides shelter unless minimal per reference condition in certain ecoregions.	
	Aquatic Invertebrate Habitat ^{C, F} : Stable substrates (cobble/wood/leaf packs) exist if expected per reference condition, (e.g., ephemeral conditions differ). Varied substrates (gravel, cobble, wood) provide multiple habitats, unless uniform is normal.	
	Lateral/Off-Channel Habitats ^{F, F} : Backwater areas are accessible as nursery/refuge unless naturally confined per reference condition.	
	Score: Notes/Other Metrics:	
Population Support: Physical and chemical conditions enable spawning, juvenile growth, migration, and stress survival of target taxa.	Fish habitat ^{C, F} : Clean gravel/vegetation is present unless alternative spawning is normal per reference condition. Sufficient cover (wood, undercuts, veg) for recruitment/survival if expected per reference condition.	
	Aquatic Invertebrate Community ^{C, F} : Diverse macroinvertebrates (incl. EPT) if expected per reference, unless ephemeral dryness limits.	
	Fish Presence ^{C, F} : Multiple native fish species if expected per reference, or unless headwaters were fishless. Juveniles/larvae appear if breeding is normal, unless ephemeral dryness excludes fish.	
	Amphibian/ Crayfish Presence ^{C, F} : Healthy numbers if expected per reference, unless ephemeral dryness excludes them per reference.	
	Score: Notes/Other Metrics:	
Community Dynamics: Species interactions sustain biodiversity, restrain invasives, and build resilience to disturbance.	Native and Non-Native Species ^{C, F} : Most historically native fish/inverts remain, unless naturally low-diversity (headwaters, ephemeral). Invasives don't dominate if native-dominated per reference, unless naturalized species cause minimal impact.	
	Species Richness, Composition, and Abundance ^{C, F} : Species richness and composition, trophic composition, fish abundance and composition; or IBI score.	
	Riparian Communities ^{F, D/F} : ~30 m or appropriate for shade/leaf input if historically expected, unless narrower is normal (e.g., prairie).	
	Keystone Species ^{C, F} : Species like beaver, unionid mussels, large crayfish persist if expected per reference, unless absent regionally. Invasives don't dominate if historically native-dominated, unless naturalized species cause minimal impact.	
	Score: Notes/Other Metrics:	
Watershed Connectivity: Continuous pathways for free movement of water, sediment, and organisms supporting recolonization, maintaining species diversity, and enabling recovery.	Upstream/Downstream Barriers ^{C/W, D} : No major barriers impede migration unless natural waterfalls exist.	
	Culvert/Road Crossing Passability ^{C, F} : Crossings allow aquatic organism passage, unless perched/undersized culverts are normal per reference.	
	Lateral Connectivity for Riparian Fauna ^{F, F} : Floodplain/wetlands accessible for semiaquatic fauna unless steep canyons preclude it.	
	Dewatered/Intermittent Segments ^{C, D} : No artificial dryness in a historically perennial system, unless ephemeral dryness is normal per reference.	
	Score: Notes/Other Metrics:	